SOLUTION REQUIREMENTS

Functional & Non-functional Requirements | Project Design Phase-II

Date	2 November	**Team ID**	_____
Project Name	Electric Vehicle Performance and Insights	**Maximum Marks**	4 Marks

Functional Requirements

The following requirements describe the Electric Vehicle Performance and Insights solution implemented in Tableau Desktop using the EV market CSV dataset.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
1	Data Access & Integration	Connect to `ev_market_train.csv` as the project data source and make the available EV market fields accessible for analysis. The workbook uses a CSV connection; no live database integration is specified.	High
2	Data Processing	Use Tableau data fields and aggregations to analyse annual sales, price, range, customer rating, manufacturer, market region, category, model year and battery type. Apply calculated fields defined for the project where required.	High
3	Business Analysis	Provide Total EV Sales (SUM of Annual Sales Units), Average EV Price (AVG of Price USD), Average Driving Range (AVG of Range Km) and Average Customer Rating (AVG of Customer Rating). Support sales contribution percentage and average sales per vehicle calculations.	High
4	Business Insights & Analytical Outputs	Provide six worksheets: EV Sales by Manufacturer, EV Sales by Region, EV Sales by Category, EV Sales by Model Year, EV Sales by Battery Type and EV Performance Analysis. Present them in the EV Market Performance Dashboard and Tableau Story.	High
5	User Interaction & Analysis	Provide Market Region and Model Year filters so users can focus the dashboard analysis. Display relevant chart information through worksheet marks and tooltips where configured.	High
6	Data Update & Reporting	Allow the workbook to be reopened and refreshed against the connected CSV source when an updated compatible file is available. Provide dashboard and story views for analysis and presentation. Automated scheduling or live refresh is not part of the confirmed implementation.	Medium

Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should use clear titles, readable chart labels, a consistent layout and straightforward Market Region and Model Year filters so users can explore the analysis with minimal confusion.
NFR-2	Security	The workbook should be shared only with intended users and stored in an appropriate location. No user authentication or role-based access feature is claimed as part of the current Tableau project.
NFR-3	Reliability	Charts and KPI views should consistently use the connected dataset and defined aggregations. The workbook should open without broken data-source references.
NFR-4	Performance	The dashboard should respond reasonably when users change filters and navigate story points. Performance should be evaluated on the target computer and dataset; no benchmark result is asserted here.
NFR-5	Availability	The workbook should be accessible wherever Tableau Desktop and the required source file are available. Availability depends on access to the workbook, source data and compatible Tableau environment.
NFR-6	Scalability	The solution structure should allow additional records or compatible fields to be analysed, subject to Tableau capacity, data quality and workbook performance. Large-scale enterprise scaling is not claimed.